MORNING BOX

Your Personal Breakfast Assistant

MB-REC-001
Recommendation & Decision Logic Standard

Version 1.0 | FINAL / LOCKED

Morning Box Product & Decision Standard

Document Purpose

This standard defines how the Morning Box MVP Recommendation Engine converts the user's Morning Profile and Daily Context into a safe, deterministic breakfast recommendation, controlled customization options and a traceable final breakfast configuration.

Morning Box recommends first, customizes second, and protects eligibility at every step.

1. Recommendation Engine Architecture

1.1 Purpose

The Recommendation Engine selects the most suitable eligible breakfast from the approved Morning Box Base Recommendation Pool.

The engine does not invent breakfasts outside the approved Morning Box Library or the Base Recommendations defined in MB-PRO-001.

1.2 Four Decision Stages

Stage	Function
Understand	Receive User Profile and Daily Context.
Filter	Apply safety, dietary and eligibility rules before preference matching.
Match	Match eligible candidates by Eating Style, Daily Context and selected Breakfast Preferences.
Recommend	Return Your Morning Box and, when available, Another Option for You.

1.3 Architecture

User Profile + Daily Context -> Safety & Dietary Filter -> Eating Style Filter -> Daily Context Match -> Breakfast Preference Pool -> Eligible Base Recommendations -> Your Morning Box + Another Option for You

Safety always comes before preference.

1.4 MVP Rule

Same Inputs -> Same Recommendation.

The MVP engine is deterministic and rule-based. It does not use random recommendations or make unsupported AI claims. Learned behavior may be added later without weakening safety rules.

2. Engine Inputs

Input	Approved values / role
Eating Style	Health-Conscious / Flexible
Breakfast Preference	Savory Breakfast / Bread & Pastry / Fresh & Wholesome / Sweet Breakfast; multi-select
Dietary Requirements & Allergies	Hard constraints used for eligibility and safety filtering
Daily Context	Active Morning / Regular Workday / Busy & On the Go / Relaxed Morning

Recommend first -> Customize second.

Size, Swap, Add-ons, Quantity, Delivery Window and Bakery are not initial recommendation inputs. They are handled after the breakfast recommendation.

3. Safety & Eligibility Filter

Base Recommendation -> Component Check -> Approved Safe Substitution -> Eligible / Ineligible

- Every component is checked against the verified Ingredient & Allergen Record.
- If a component is incompatible, the engine first checks for an approved compatible substitution.
- A substitution may only come from the Morning Box-approved Component Library, never the Partner Bakery's general menu.

- If no approved compatible substitution exists, the Base Recommendation is removed from the Candidate Pool.

Safety requirements are never relaxed to create a recommendation.

If no approved compatible breakfast remains, Morning Box returns no recommendation rather than creating an unapproved substitute.

4. Eating Style Logic

4.1 Health-Conscious

The engine normally evaluates only Health-Conscious Base Recommendations. Savory proteins such as Chicken Sausage, Beef Bacon and Turkey Bacon are not default recommendations in this branch. Standard pastries are also not default recommendations for Health-Conscious users. Health-Conscious is a recommendation direction, not a safety prohibition.

4.2 Flexible

The Flexible branch may use the broader approved Core Library, including Flexible-only savory proteins and standard pastries, after all safety and dietary filters.

4.3 No Automatic Cross-over

Health-Conscious and Flexible remain separate recommendation branches. The engine does not silently cross from one branch to the other merely to create a result.

> ***Eating Style defines the recommendation branch, while Safety defines eligibility. Eating Style may guide what Morning Box recommends, but it never overrides dietary or allergy requirements.***

5. Daily Context Matching Logic

Daily Context belongs to the specific breakfast day, not the permanent Base Morning Profile. Different days may use different contexts without changing the user's core profile.

> ***Daily Context is respected, not silently replaced.***

The engine matches within the selected Active Morning, Regular Workday, Busy & On the Go or Relaxed Morning context. If a candidate becomes ineligible, the engine may consider other eligible selected Preferences in the same context, but it does not silently switch context.

> ***Daily Context defines the breakfast situation for that specific day. The Recommendation Engine matches within the selected Daily Context and does not silently switch context to create a recommendation.***

6. Breakfast Preference Pool Logic

If one Breakfast Preference is selected, the Candidate Pool is limited to that Preference within the selected Eating Style and Daily Context. If multiple Preferences are selected, each creates an eligible candidate path; categories are never merged into a hybrid breakfast.

> ***Multiple preferences expand the recommendation pool; they do not combine multiple breakfast categories into one box.***

> ***Breakfast Preferences define the eligible recommendation pool. Selecting multiple preferences expands that pool without combining categories or implying preference ranking.***

The order in which a user taps multiple Preferences does not create a ranking. Tie-breaking is handled by the context-based priority rules in Section 7.

7. Recommendation Priority & Tie-Breaking Logic

Context Priority is used only when the user selected multiple Breakfast Preferences and more than one eligible candidate remains. It never introduces a Preference the user did not select.

Daily Context	Priority 1	Priority 2	Priority 3	Priority 4
Active Morning	Savory	Fresh & Wholesome	Bread & Pastry	Sweet
Regular Workday	Savory	Fresh & Wholesome	Bread & Pastry	Sweet
Busy & On the Go	Bread & Pastry	Savory	Sweet	Fresh & Wholesome
Relaxed Morning	Savory	Bread & Pastry	Fresh & Wholesome	Sweet

7.1 Active Morning

Savory is first because the approved Active Savory models provide the most complete active-morning structure; Fresh & Wholesome follows with oats/yogurt/grain/fruit structures. Bread & Pastry and Sweet remain valid when selected.

7.2 Regular Workday

Savory and Fresh & Wholesome receive priority as balanced everyday workday structures, followed by Bread & Pastry and Sweet.

7.3 Busy & On the Go

For Busy & On the Go, portability and ease of eating determine the tie-break priority.

Bread & Pastry is first, followed by ready-to-eat Savory, then Sweet. Fresh & Wholesome remains valid but is lower in a tie because bowls/cups may require more handling.

7.4 Relaxed Morning

For a Relaxed Morning, completeness and breakfast experience take priority over portability and speed.

Savory is first, followed by Bread & Pastry, Fresh & Wholesome and Sweet.

8. "Another Option for You" Logic

Your Morning Box = Highest-priority eligible candidate

Another Option for You = Next highest-priority eligible candidate

The second option must remain within the user's selected Breakfast Preferences, the same Eating Style and the same Daily Context, while satisfying all safety and dietary rules. If only one eligible candidate exists, only Your Morning Box is shown.

Another Option for You is the next highest-priority eligible recommendation within the user's selected preferences and the same Eating Style and Daily Context. If no meaningful eligible alternative exists, no second recommendation is shown.

9. Swap Eligibility Logic

Recommendation -> Component -> Swap Group -> Eligible Alternatives

- Swap is post-recommendation and does not determine the initial Your Morning Box.
- Before an alternative is shown, Safety, Dietary and Compatibility filters are applied again.
- Swap cannot bypass the user's Eating Style branch.
- No Direct Swap items from MB-PRO-001 - Avocado, Olives, Greek Yogurt and Overnight Oats - do not display a Swap action.
- Each replacement uses its own approved portion standard; the role is preserved, not exact weight or nutrition.
- Approved Swap within the same Swap Group causes no price change.

Swap is a controlled post-recommendation customization. It may only use eligible alternatives from the approved Swap Group and must never bypass Safety, Dietary, Eating Style or product compatibility rules.

10. Dietary Substitution vs User Swap

Change type	When	Purpose
System Substitution	Before recommendation is shown	Make an otherwise compatible Base Recommendation eligible using an approved safe/dietary substitute.
User Swap	After recommendation is shown	Allow controlled user customization from eligible alternatives.

Both change types remain inside the Morning Box-approved Library and all safety, dietary and compatibility rules. Backend records the origin as System Substitution or User Swap.

System Substitution makes a Base Recommendation eligible before it is shown. User Swap allows controlled customization after the recommendation is shown. Both must remain within the approved Morning Box Library and all applicable safety and compatibility rules.

11. Size Selection Logic

Recommendation -> Swap (if desired) -> Size -> Add-ons

- Size is post-recommendation and never causes a different breakfast to be recommended.
- Where applicable: Light / Regular / Large, with Regular selected by default.

Size changes quantity, not composition.

- Each component follows its own approved MB-PRO-001 portion rule; fixed-unit components such as pastries remain fixed where specified.
- For very simple Busy & On-the-Go breakfasts where size variation creates no meaningful product difference, Morning Box may offer one Standard Size only and show no Size Selector.

Size is a post-recommendation quantity adjustment. It follows each component's approved portion specification and never changes the composition of the recommended breakfast. Where size variation provides no meaningful product difference, Morning Box may offer one Standard Size only.

12. Add-on Logic

Recommendation -> Swap -> Size -> Add-ons -> Confirm

Add-ons are optional and do not influence Your Morning Box or Another Option for You.

MVP Add-on Group	Allowed options
Bread Add-ons	White Sourdough / Whole Wheat Sourdough / Multigrain / Ciabatta / Focaccia -> Half Loaf or Whole Loaf -> Quantity
Pastry Add-ons	Plain Croissant / Cheese Croissant / Pain au Chocolat / Fruit Danish -> Quantity

- No other Core components are standalone Add-ons in the MVP.
- The Add-on Library is filtered through applicable Safety and Dietary rules before display.
- If an Add-on Group has no eligible products, that group is not shown.

Default = No Add-on.

Add-ons are optional post-recommendation extras and do not influence the breakfast recommendation. Only approved and eligible Bread and Pastry Add-ons are shown, after all applicable safety and dietary filters have been applied.

13. No-Match & Fallback Logic

No Eligible Candidate -> No Recommendation

- The engine does not weaken Safety, Dietary Requirements, Eating Style or Daily Context merely to create a result.
- If only one eligible candidate exists, show Your Morning Box and do not show Another Option for You.

One strong recommendation is better than two forced recommendations.

- If a swappable component has no eligible alternative after filtering, do not show Swap.
- If an Add-on Group has no eligible item after filtering, do not show that group.

13.1 User-facing No-Match Message

We couldn't find a Morning Box that safely matches your requirements.

Action: Review my preferences

The user may review their own inputs. The system does not automatically relax them.

Morning Box never weakens safety, dietary requirements, Eating Style or Daily Context simply to produce a result. When no eligible option exists, the system returns no recommendation and gives the user a simple path to review their inputs.

14. Learning & Feedback Logic

After breakfast, the MVP may collect two simple feedback signals: Loved it and Not for me.

- Loved it = positive preference signal.
- Not for me = negative preference signal; it is not an allergy or dietary restriction.

Safety Data != Preference Data

The backend should associate feedback with the User, Base Recommendation, final breakfast after System Substitution/User Swap, Daily Context and feedback value.

Collect feedback first. Do not automatically change deterministic recommendation rules based on a single feedback event.

Morning Box records feedback as a learning signal without allowing feedback to override safety or dietary rules. MVP recommendations remain deterministic; behavioral learning may influence future versions once sufficient reliable data exists.

15. Recommendation Output & Backend Record

Each recommendation must be traceable from the inputs to the final breakfast configuration. This decision detail is backend-facing and should not add complexity to the customer experience.

Record	Minimum backend data
Inputs	Eating Style; Selected Breakfast Preference(s); Daily Context; Dietary/Allergy Filters applied
Recommendation	Selected Base Recommendation; Your Morning Box; Another Option for You if available
Changes	System Substitutions; User Swaps; Selected Size or Standard Size; Add-ons
Output	Final Breakfast Configuration

15.1 Internal Recommendation Reason Codes

Internal reason codes may include PRIMARY_CONTEXT_PRIORITY, SECONDARY_ELIGIBLE_OPTION, SYSTEM_DIETARY_SUBSTITUTION and NO_ELIGIBLE_RECOMMENDATION. These codes are for development, debugging and analytics, not customer-facing UI.

Every recommendation must be traceable from user inputs to the final breakfast configuration, while backend decision complexity remains invisible to the customer.

16. Final End-to-End Logic Audit

Base Morning Profile -> Eating Style + Breakfast Preferences + Dietary Requirements & Allergies -> Daily Context -> Safety & Dietary Eligibility -> Eating Style Branch -> Selected Preference Pool -> Exact Daily Context Match -> Approved System Substitution if required -> Eligible Base Recommendations -> Context Priority / Tie-Breaking -> Your Morning Box -> Another Option for You if eligible -> User Swap optional -> Size (Regular default / Standard Size where applicable) -> Add-ons optional -> Final Breakfast Configuration -> Order -> Feedback

16.1 Final Engine Rules

- 1. Safety always comes first.
- 2. Only Morning Box-approved products and Base Recommendations may be used.
- 3. Same Inputs -> Same Recommendation in MVP.
- 4. Eating Style defines the recommendation branch.
- 5. Daily Context is never silently changed.
- 6. Multiple Breakfast Preferences expand the pool; they are never combined into one hybrid breakfast.
- 7. Context Priority resolves ties between multiple eligible selected Preferences.
- 8. Customization happens after Recommendation and cannot bypass eligibility rules.
- 9. No eligible match is better than an unsafe or forced recommendation.
- 10. Feedback is collected for learning but does not override deterministic MVP rules.

Morning Box recommends first, customizes second, and protects eligibility at every step.

Final Status

MB-REC-001 - Recommendation & Decision Logic Standard - Version 1.0

FINAL / LOCKED

This document consolidates all finalized Morning Box MVP recommendation and decision logic decisions through Sections 1-16.